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Oefeningenreeks 2D nr. 11.8

$$\begin{aligned} & \lim_{x \rightarrow 9} \left( \frac{\sqrt{3x-2} - \sqrt{2x+7}}{x-9} \right) \\ &= \lim_{x \rightarrow 9} \left( \frac{\sqrt{3x-2} - \sqrt{2x+7}}{x-9} \cdot \frac{\sqrt{3x-2} + \sqrt{2x+7}}{\sqrt{3x-2} + \sqrt{2x+7}} \right) \\ &= \lim_{x \rightarrow 9} \left( \frac{(3x-2) - (2x+7)}{(x-9)(\sqrt{3x-2} + \sqrt{2x+7})} \right) \\ &= \lim_{x \rightarrow 9} \left( \frac{x-9}{(x-9)(\sqrt{3x-2} + \sqrt{2x+7})} \right) \\ &= \lim_{x \rightarrow 9} \left( \frac{1}{\sqrt{3x-2} + \sqrt{2x+7}} \right) \\ &= \frac{1}{\sqrt{25} + \sqrt{25}} \\ &= \frac{1}{5+5} \\ &= \frac{1}{10} \end{aligned}$$

## References